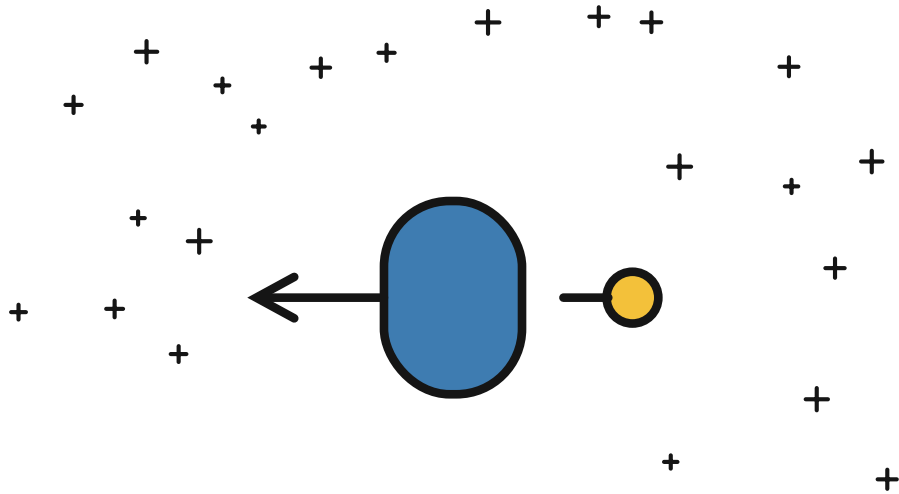


Next:

Push Against Nothing

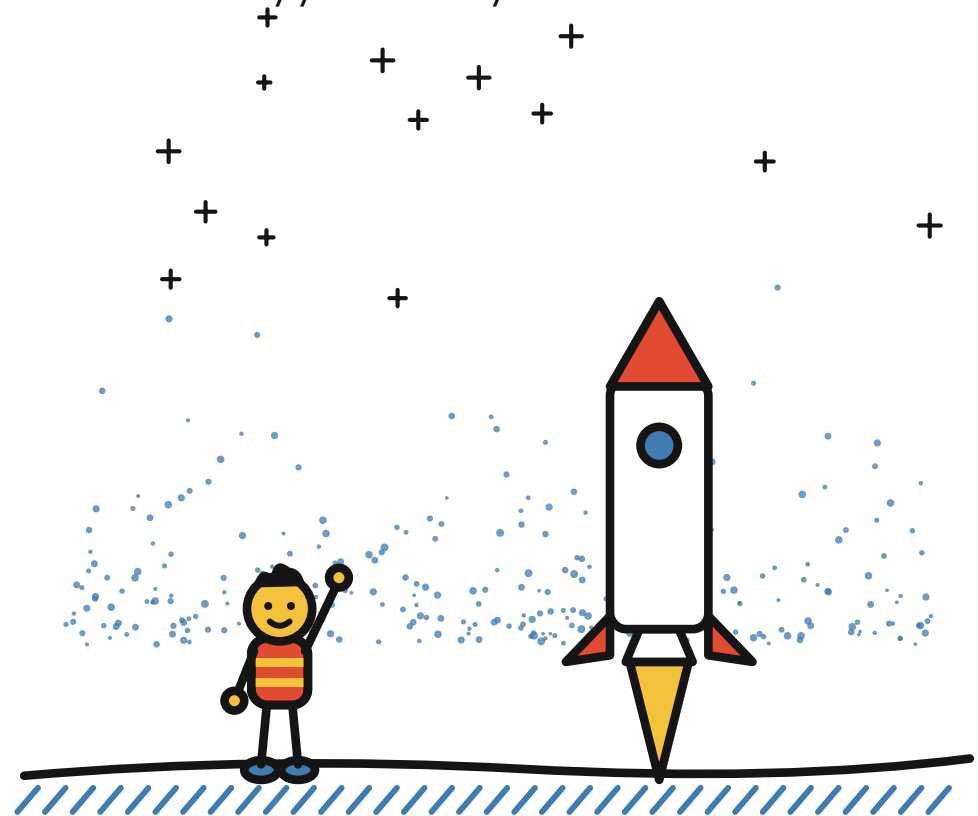
Book Three · how you move where there is nothing to push



Throw something backwards
and you go forwards.

THE AIR RUNS OUT

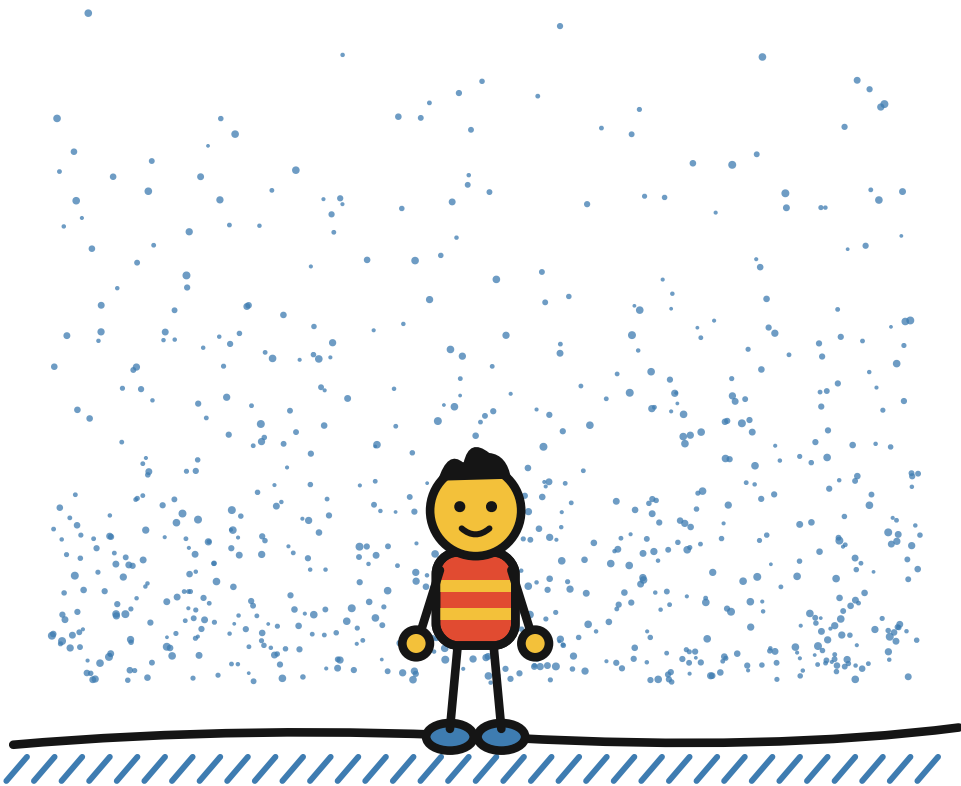
Book Two · Why you cannot fly to the Moon



Print it. Read it together. Then light a candle.

You are breathing.

In. Out. In. Out. All day long.



That is air. Down here, it is everywhere.

For the grown-up.

The one idea

A rocket is not defined by its shape. It is defined by the fact that it carries its own oxidiser. Everything pointy and finned is incidental; the tank of oxygen is what makes it a rocket rather than a very fast plane.

Why the ceilings are real

Air density halves roughly every 5.5 kilometres. At 20 km there is about 7% of sea-level air; by 50 km, about a thousandth. Wings stop working and air-breathing engines starve. The SR-71 topped out near 26 km. The highest crewed balloon flight reached about 41 km.

The counter-intuitive part

Because it hauls its own oxygen, a rocket is far less efficient than a jet — roughly two-thirds of the propellant mass on a launch vehicle is oxidiser, not fuel. We accept that enormous penalty because there is no alternative above the air.

If he asks where space starts

There is no edge. The 100 km Kármán line is a convention. NASA and the US military have often used 80 km instead. The air simply thins until it stops mattering.

Try this tonight.

1 Put a jar over a candle.

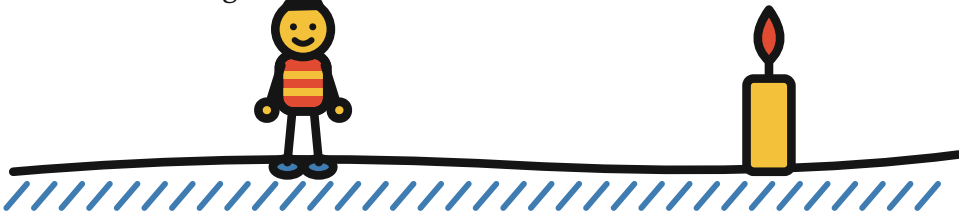
Light a short candle, lower a glass jar over it, and count together. It goes out in seconds. Ask him where the air went — that is the whole book in one answer.

2 Fan a piece of paper.

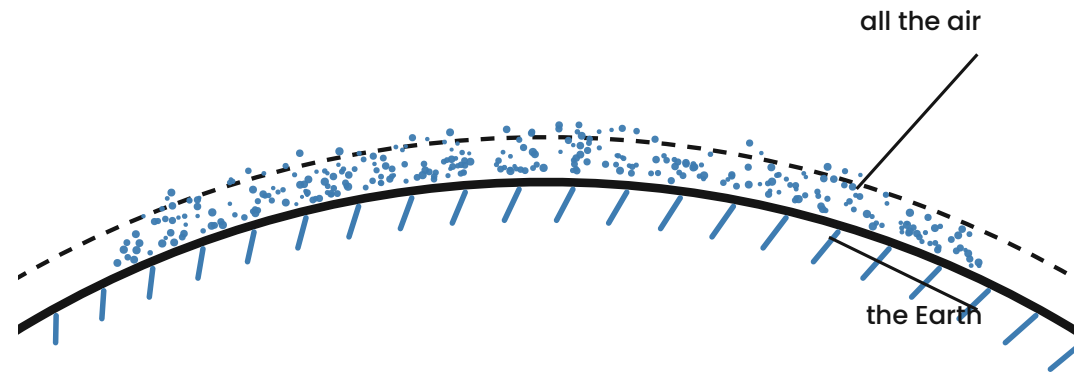
Wave a sheet of paper and feel it push back. That push is what a bird's wing and a plane's wing both live on. Take the air away and there is nothing to push.

3 Blow up a balloon and squeeze it.

The air inside is packed tight. That is what a rocket carries — its own air, squeezed small enough to bring along.

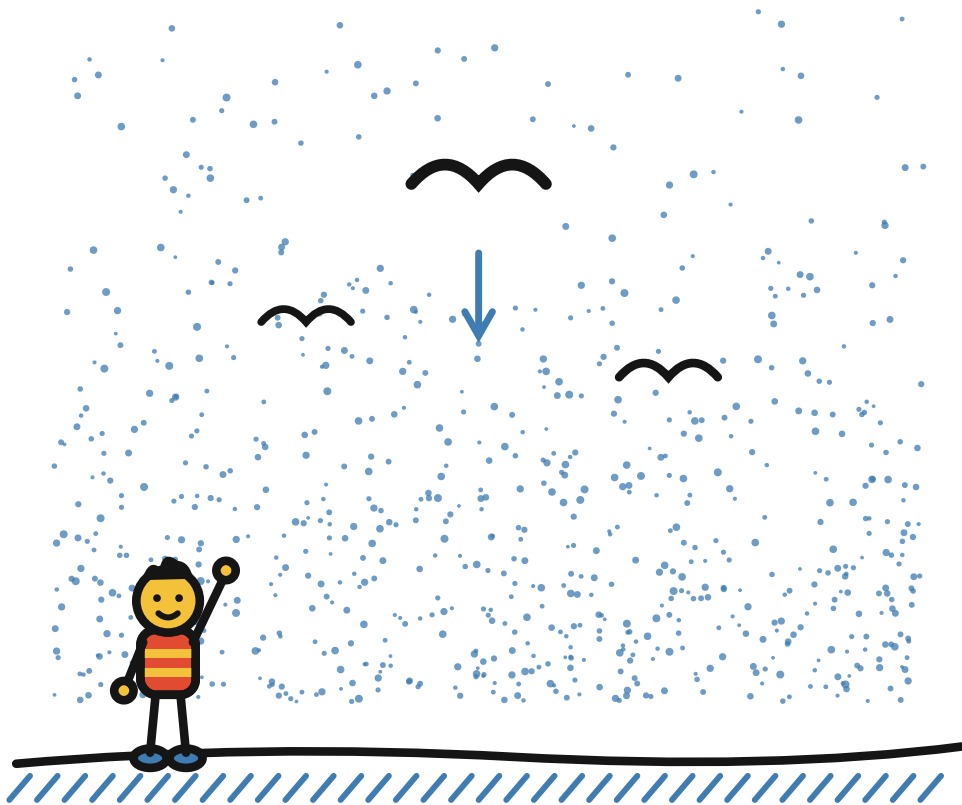


But there is not
very much of it.



A thin coat around a very big ball.

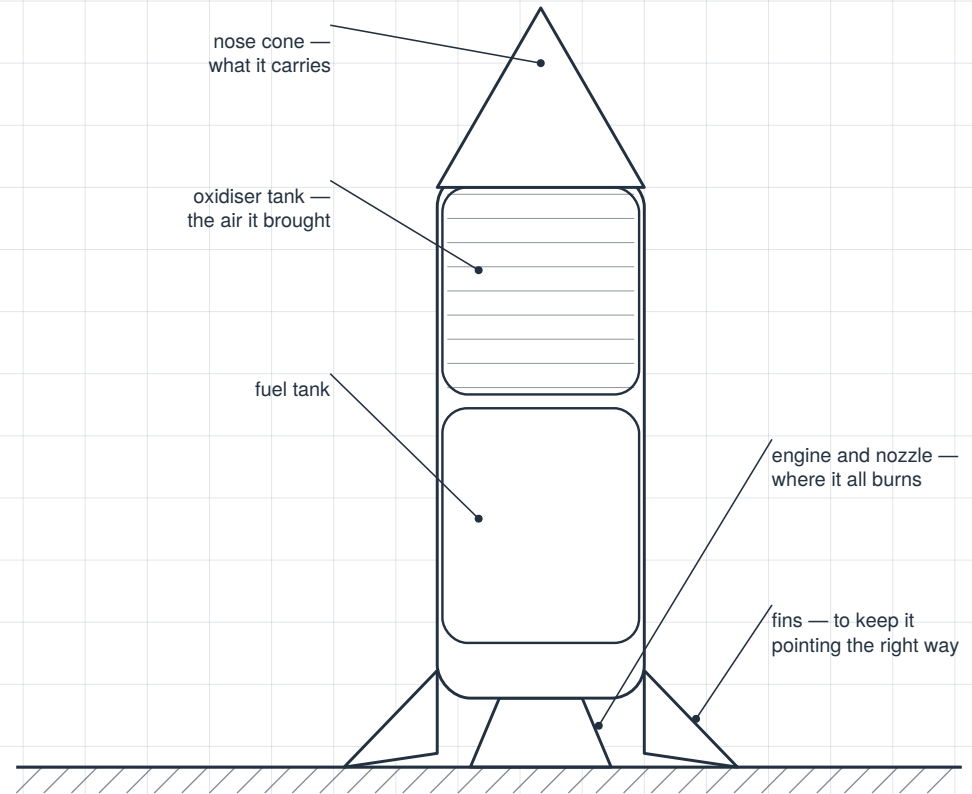
A bird pushes on the air.



It pushes air down, so it goes up.

Inside a rocket.

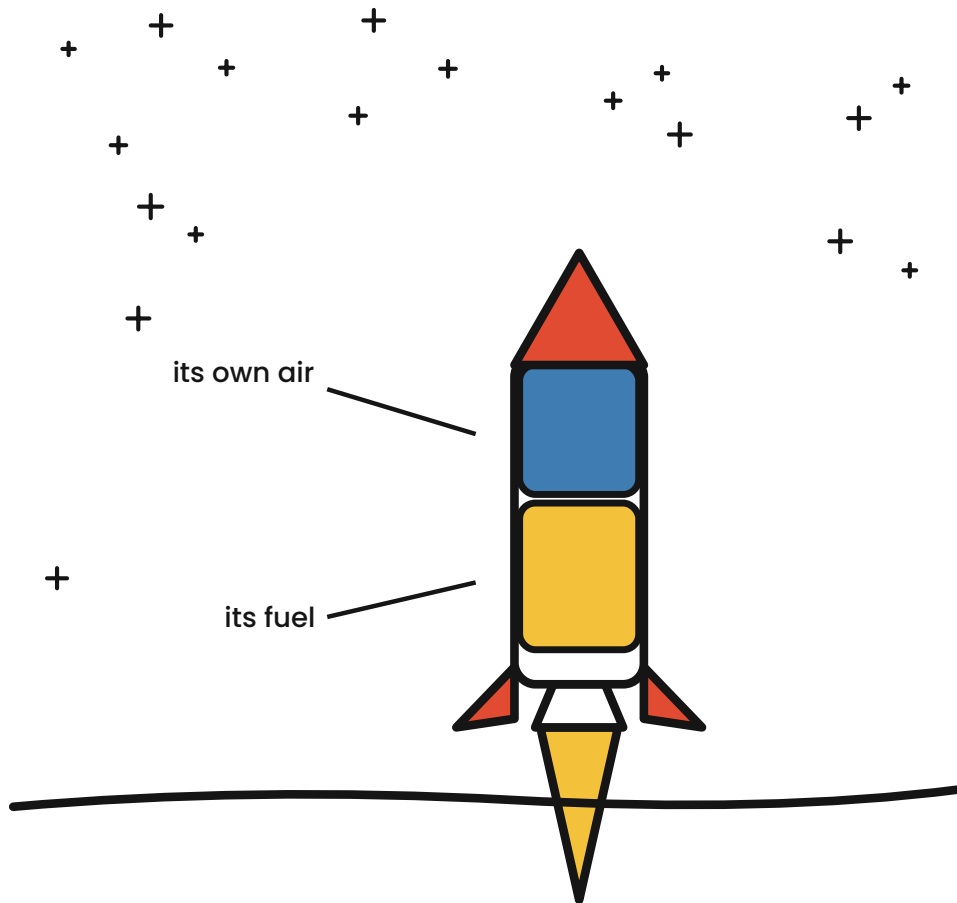
The Air Runs Out · figure 2 · the parts



Two tanks. One holds the fire; the other holds what the fire needs.

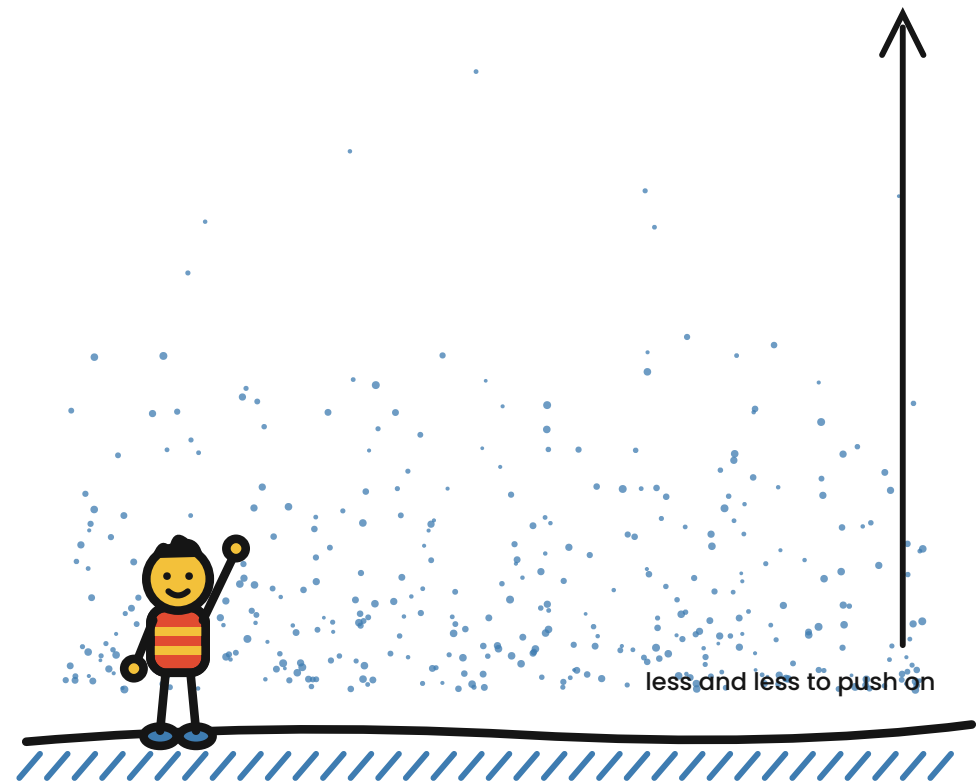
That is a rocket.

Fuel to burn, and air to burn it with.

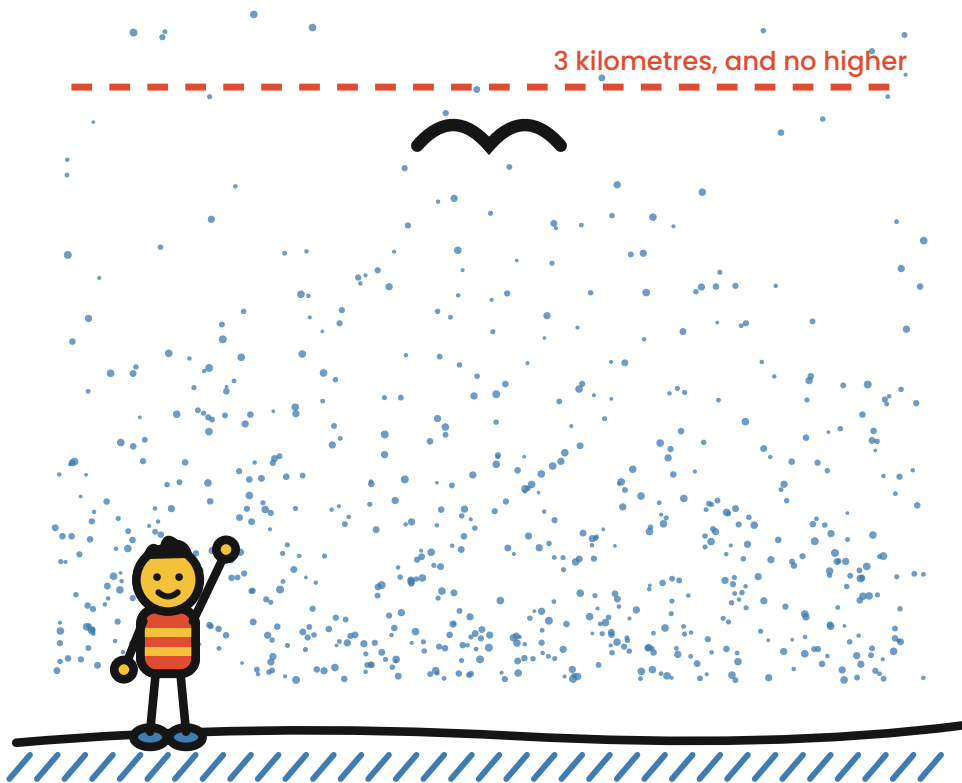


It carries the sky it needs.

Go up, and the air gets thinner.



**So a bird can only
go so high.**



Above that, there is not enough air to flap.

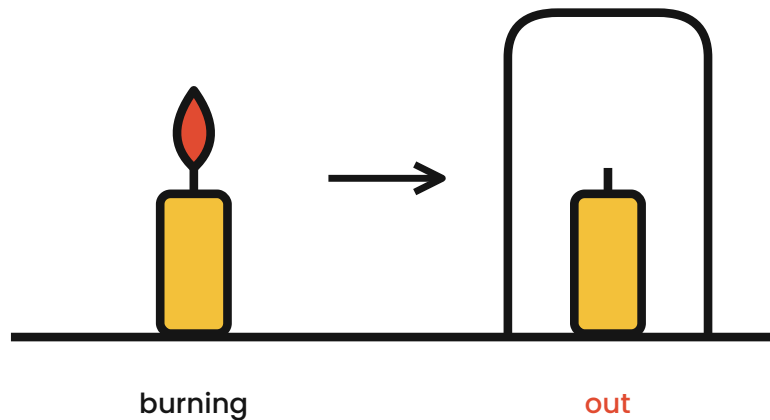
**So what if you
brought your own?**



Then the empty sky would not matter at all.

Fire needs air too.

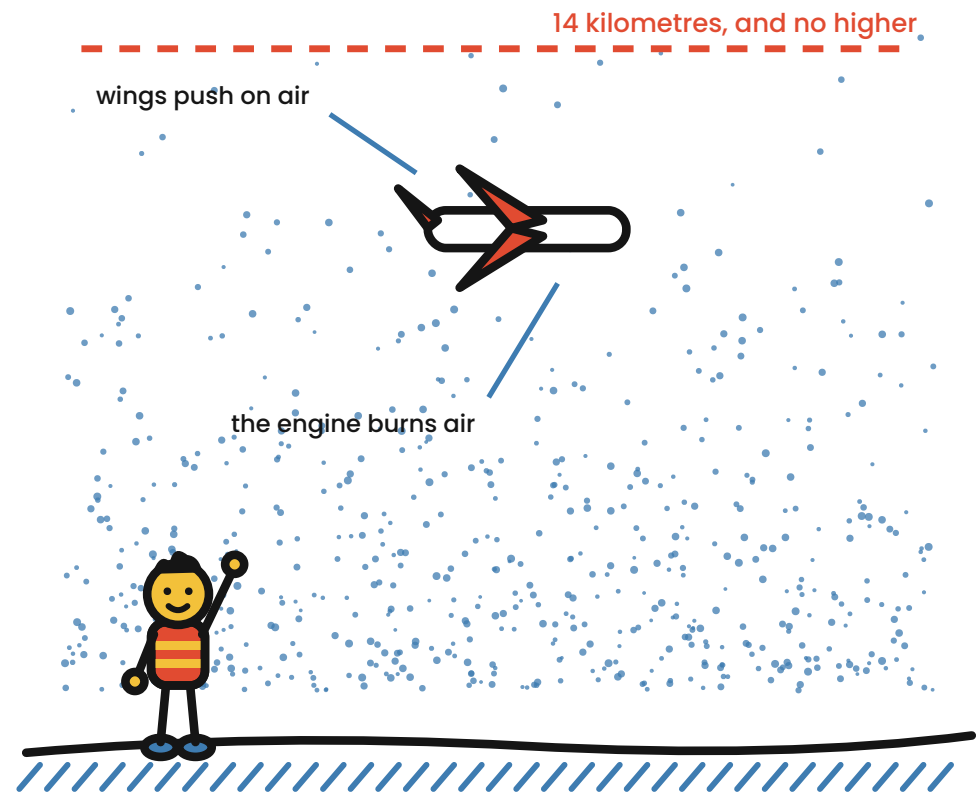
Watch what happens.



The jar ran out of air. So did the flame.

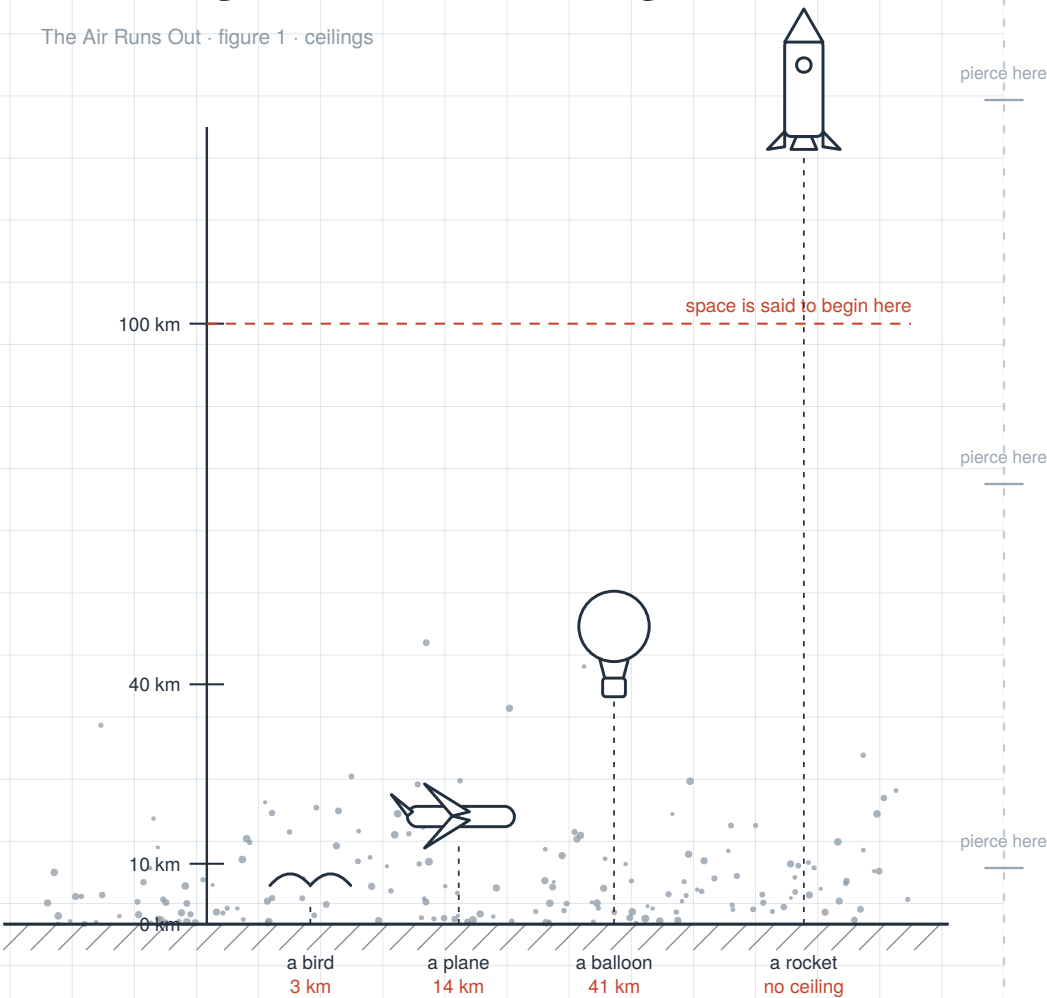
A plane goes higher.

But it needs air twice over.



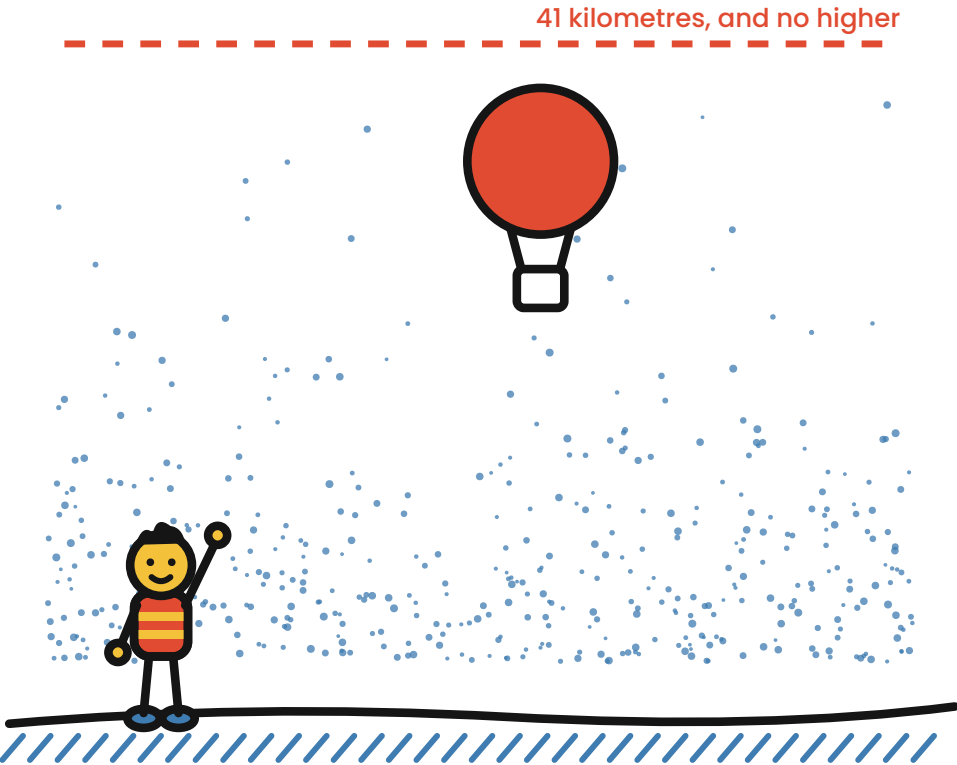
How high can each one go?

The Air Runs Out · figure 1 · ceilings



Three of them run out of air. One brought its own.

A balloon floats on the air.



Where there is no air, there is nothing to float on.